

BALAJI N

Senior Electronics Engineer

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SUMMARY

Electronics Service Engineer with 9+ years of hands-on experience in Apple device repair and PCB-level troubleshooting, including MacBook, iPhone, and other portable electronics. Highly skilled in SMD soldering, de-soldering, and component-level fault diagnosis. Strong expertise in identifying and resolving complex hardware issues using systematic troubleshooting and circuit analysis techniques, including UPS and power electronics domain, leveraging strong fundamentals in electronics and repair.

EXPERIENCE

06/2016 - 12/2026

Chennai,Tamil Nadu

PROFESSIONAL EXPERIENCE

- **Service Engineer – Apple Devices & Electronics Repair Tring Ring Mobile Service, Chennai June 2016 – Feb 2026.**
- **MacBook Motherboard Repair.**
 - Identified faulty capacitor causing no power issue
 - Replaced component and restored system functionality
- **UPS / Inverter Repair.**
 - Hands-on experience in UPS rectification and servicing (1kVA – 10kVA range).
 - Worked on multiple brands including BPE, ProstarM, Microtek, and Delta UPS systems
- **iPhone PCB Micro-Soldering.**
 - Replaced damaged IC using hot air station
 - Restored full device functionality
- **Laptop Charging Circuit Repair.**
 - Diagnosed MOSFET failure in charging section
 - Replaced components and validated charging stability

2016 - 2026

Chennai,Tamil Nadu

TECHNICAL EXPERTISE IN TROUBLESHOOTING

- **Component-level fault identification.**
- **Short circuit detection and isolation.**
- **Voltage rail analysis , Power section diagnostics.**
- **Signal tracing in compact PCBs ,Root cause analysis for repeated failures.**

KEY ACHIEVEMENTS

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Major Achievements.
 - Successfully handled complex motherboard repairs with high accuracy.
 - Reduced repeat failures by ensuring root cause identification.
 - Built strong customer trust through reliable repair solutions.
 - Managed high-volume repairs with consistent quality output.
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15% Conversion Rate Boost
 - Diagnosed and repaired MacBook, iPhone, iPad hardware issues at board level
 - Performed SMD soldering and micro-level component replacement
 - Identified faults in power circuits, charging sections, display and signal lines
 - Repaired damaged PCBs including burnt components, broken tracks, short circuits
 - Used systematic troubleshooting methods: voltage testing, continuity checks, fault isolation
 - Installed OS, performed firmware updates, and optimized device performance.

EDUCATION

- Bachelor of Engineering (B.E.) – Electronics and Communication Engineering.

LANGUAGES

Tamil
Native

English
Proficient

CORE TECHNICAL SKILLS

UPS & Power Electronics Experience

- Hands-on experience in UPS rectification and servicing (1kVA – 10kVA range)
- Worked on multiple brands including BPE, ProstarM, Microtek, and Delta UPS systems. •Diagnosed and repaired power section faults, charging issues, and inverter stage failures.
- Performed component-level troubleshooting including MOSFETs, IGBTs, capacitors, and driver circuits.
- Identified and resolved issues like no output, overload, battery not charging, and shutdown faults.
- Experience in SMPS and DC bus voltage analysis.
- Conducted testing, validation, and performance checks after repair.

PCB & Repair Skills

- SMD Soldering & De-soldering (Hot air rework station).
- Through-hole component replacement.
- PCB track repair and jumper techniques.
- Micro-soldering for compact boards.
- Board-level diagnostics and repair.

Electronic Components Knowledge.

- Resistors, Capacitors (Electrolytic, Ceramic)
- Diodes (Zener, Rectifier)
- Transistors (BJT, MOSFET – basic understanding)
- ICs (Power ICs, Logic ICs, Driver ICs)
- Connectors, Coils, Filters

Device Expertise

- MacBook hardware repair
- iPhone board-level troubleshooting
- Laptop motherboard diagnostics
- Battery and charging circuit repair,

Tools & Equipment.

- Digital Multimeter
- Soldering Station & Hot Air Gun
- DC Power Supply
- Microscope for SMD work
- Basic Oscilloscope usage